

RAZI MAHMOOD

Trustworthy Multimodal AI Researcher | Medical Imaging, Computer Vision, NLP | Ph.D. Student at UBC

raziuddin.mahmood@gmail.com | (408) 540-4031 | razi-mahmood.github.io | [LinkedIn](#) | [GitHub](#) | U.S. Citizen

RESEARCH AND ENGINEERING PROFILE

Research-oriented data scientist and Biomedical Engineering Ph.D. student at the University of British Columbia with a publication-backed record in multimodal AI, machine learning, deep learning, NLP, computer vision, model evaluation, and applied ML systems. Passionate about developing AI methods for real-world problems with societal impact, with recent work spanning trustworthy radiology report verification, head CT report generation, large-scale image-text datasets, RAG/vector search evaluation, data annotation workflows, Slurm-based model training, and CI/CD testing for ML systems.

EDUCATION

Ph.D., Biomedical Engineering, University of British Columbia

2026 - Present

Vancouver, BC

- Advisor: Prof. Purang Abulmousseli. Research interests include multimodal AI, computer vision, medical imaging, NLP, text analytics, and ML systems.

M.S., Biomedical Engineering, Rensselaer Polytechnic Institute

2023 - 2026

Troy, NY

- Master's thesis: Image-Driven Fact-Checking of AI Generated Chest Radiology Reports; nominated for Best Master's Thesis Award.
- Graduate researcher in the Deep Image Analysis Lab under Dr. Pingkun Yan, with clinical guidance from Dr. Mannudeep Kalra.

B.A., Data Science, Domain Emphasis in Cognition, University of California, Berkeley

2018 - 2022

Berkeley, CA

RESEARCH AND WORK EXPERIENCE

Machine Learning / AI Research Intern, HOPPR

2026

- Developed a head CT radiology report generation model using fine-grained finding prediction and probabilistic patient-similarity retrieval.
- Built a non-LLM generation approach designed to improve clinical grounding, report accuracy, and hallucination performance relative to LLM-based generation baselines.
- Worked across multimodal medical AI, computer vision, radiology text generation, retrieval, and generated-report evaluation.

Graduate Researcher, Deep Image Analysis Lab, Rensselaer Polytechnic Institute

May 2023 - June 2025

- Developed discriminative fact-checking models to verify whether AI-generated chest X-ray reports are clinically consistent with medical images.
- Built image-driven and phrase-grounded methods for evaluating and correcting generated radiology reports, showing 15-40% quality improvement in experimental settings.
- Created and open-sourced RadCheck, a large-scale dataset of real/fake image-finding pairs for radiology report verification on Hugging Face.
- Ran large-scale model training and evaluation workflows on Slurm-managed GPU clusters, including distributed experiments for medical image-text modeling.
- Published work across MICCAI, ISBI, MLMI/MICCAI, NeurIPS GenAI4Health, and CVPR; one patent filed.

Data Science Intern, IBM Storage Systems

July 2025 - October 2025

- Built search evaluation tooling for a content-aware RAG storage product, including benchmarks for file search, vector retrieval, and re-ranking.
- Integrated search evaluation benchmarks into daily regression and CI/CD testing workflows used by developers, performance testers, and QA automation teams.
- Developed LexLogit, a re-ranking approach combining semantic, lexical, and re-ranker signals in vector databases.

Machine Learning Research Intern, IBM Almaden Research

Sept. 2021 - April 2022; June 2014 - July 2016

- Co-developed a spatially preserving CNN architecture for location-aware chest radiograph abnormality classification, improving AUC by 10% on MIMIC data.

- Contributed to medical imaging AI projects for chest radiographs, cardiac aneurysm detection, and dilated cardiomyopathy detection; published at ISBI/AMIA and contributed to patent disclosures.

Machine Learning Research Intern, IBM Watson

Jan. 2021 - May 2021

- Developed ContrastBERT, a log anomaly classification model combining BERT-encoded IT logs with supervised contrastive learning.
- Collaborated with IBM Watson AIOps researchers on NLP/text analytics for IT operations; published at IEEE BigData-IT.

Data Science Intern, Hyperfine

Aug. 2021 - Nov. 2021

- Built NLP-based data annotation workflows for radiology text reports, extracting 7,200 annotations from 600 brain MRI reports.
- Created a Jupyter-based labeling interface that reduced manual annotation time by 10x.

Data Science Intern, Xoran Technologies

June 2021 - Aug. 2021

- Developed 3D U-Net segmentation models for cone-beam CT head and neck anatomy, achieving Dice coefficient 0.68 across nine anatomical structures.
- Supported clinical data labeling workflows and trained staff on ITK-SNAP and 3D Slicer annotation tools.

SELECTED TECHNICAL SKILLS

AI / ML	Multimodal AI, computer vision, medical imaging AI, NLP/text analytics, deep learning, vision-language models
Implementation	Python, PyTorch, TensorFlow/Keras, Hugging Face, scikit-learn, NumPy, pandas, OpenCV, SimpleITK, spaCy
ML Systems	MLOps, model evaluation, CI/CD testing, regression testing, vector search/RAG, distributed training, Slurm, GPU workflows, Git/GitHub, Docker, VS Code, Jupyter, Claude-assisted development

PUBLICATIONS

- [1] R. Mahmood and T. F. Syeda-Mahmood, "Phrase-grounded APO for Improving Chest X-ray Report Generation," accepted for CVPR 2026.
- [2] R. Mahmood, P. Yan, and T. F. Syeda-Mahmood, "Automatic Correction of AI Reports using Fact-Checking Model-guided LLMs," NeurIPS GenAI4Health Workshop, 2025; also accepted for IEEE ISBI 2026.
- [3] T. Syeda-Mahmood, K. C. L. Wong, S. Kashyap, N. D'Souza, L. Shi, R. Mahmood, and H. Wang, "SemCLIP: A Semantic Memory-Aligned Vision-Language Model," PMLR / NeurIPS UniReps, 2025-2026.
- [4] R. Mahmood, K. C. L. Wong, D. M. Reyes, N. D'Souza, L. Shi, J. Wu, P. Kaviani, M. Kalra, G. Wang, P. Yan, and T. Syeda-Mahmood, "Anatomically-Grounded Fact Checking of Automated Chest X-ray Reports," MICCAI 2025.
- [5] R. Mahmood, P. Yan, D. M. Reyes, G. Wang, M. Kalra, P. Kaviani, J. T. Wu, and T. Syeda-Mahmood, "Evaluating Automated Radiology Report Quality through Fine-Grained Phrasal Grounding of Clinical Findings," IEEE ISBI 2025.
- [6] R. Mahmood and T. Syeda-Mahmood, "Improving Open Vocabulary Tagging using Semantic Clustering," BayLearn 2024.
- [7] R. Mahmood, D. Machado, G. Wang, M. Kalra, and P. Yan, "Fact-checking of AI-generated Reports," MLMI Workshop at MICCAI, Springer Nature, 2023.
- [8] R. Mahmood, X. Liu, A. Xu, and R. Akkiraju, "Contrast BERT: Supervised Contrastive Learning of BERT-Encoded IT Logs for Anomaly Classification," IEEE BigData-IT, 2022.
- [9] N. Shrivastava, R. Mahmood, and T. Syeda-Mahmood, "Spatially-preserving Flattening in Deep Learning for Location-aware Classification," IEEE ISBI 2022.
- [10] R. Mahmood, Q. Wang, and T. Syeda-Mahmood, "Automatic Detection of Dilated Cardiomyopathy in Cardiac Ultrasound Videos," AMIA 2014.
- [11] R. Mahmood and T. Syeda-Mahmood, "Automatic Detection of Left Ventricular Aneurysms in Echocardiogram Videos," IEEE ISBI 2015.

Patents

- [P1] R. Mahmood and T. Syeda-Mahmood, "Improving Open Vocabulary Tagging using Semantic Clustering," IBM patent filed 2025, Ref: P202500951US02.
- [P2] N. Shrivastava, R. Mahmood, and T. Syeda-Mahmood, "Spatially-preserving Flattening in Deep Learning for Location-aware Classification," patent filed 2022, Ref: P202200366US01.

AWARDS, SERVICE, AND TEACHING

UC Berkeley Data Science Changemaker Award , UC Berkeley Data Science Undergraduate Studies	2026
• Recognized for M.S. thesis research on clinical reliability and trustworthy AI for societal benefit.	
Best Master's Thesis Award Nomination , RPI / Northeastern Association of Graduate Schools	2026
• Master's thesis nominated by RPI for Best Thesis Award consideration.	
UC Berkeley Undergraduate HCE Essay Prize, Runner-up	2020
• Recognized for original research at the intersection of data science, technology, and society.	
Co-organizer , MICCAI Tutorial on Foundational Models for Healthcare	2025; 2026
Co-organizer , MICCAI Workshop on Multimodal Learning for Clinical Decision Support	2025
Peer Reviewer , NeurIPS, CVPR, IEEE EMBC, ICLR, MICCAI, ML-CDS and related workshops	2024 - 2026
Teaching and Mentorship , RPI Engineering; UC Berkeley Data Science Society	2021 - 2025